PRECALCULUS
POLYNOMIALS: THEIR ROOTS AND GRAPHS

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1. Polynomial Functions

Definition. Let n be a fixed non-negative integer (i.e. $n \geq 0$ is an integer). A **polynomial of degree** n (in one variable) is a function f of the form

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0$$

where a_i is a constant for each i between $0 \leq i \leq n$.

- the integer n is called **degree**; it is always the highest exponent that appears on the variable
  - the notation: $\deg(f) = n$
- the constants a_i are called **coefficients**
- if all of the a_i are real numbers, then we say that $f(x)$ is a *polynomial with real coefficients* (this can be denoted by $f \in \mathbb{R}[x]$ in fancy maths symbols)
- the a_i just need to be *constants*, as opposed to variables. For example,

$$2x^2 + 2 - i$$

is a degree 2 polynomial but since $(2-i)$ is not a real number, but a complex one, cannot say that $2x^2+2-i$ is a polynomial with real coefficients (instead it is complex coefficients)

Definition. Let $f(x)$ be a polynomial. A number λ that satisfies the equation

$$f(\lambda)=0$$

is called a **zero** (or **root**) of $f(x)$.

- The zeros/roots of a polynomial are the x-intercepts.

Example 1. Find all of the zeros of the function

$$f(x)=4(x-2)(x+3)(x-2021)^{19}.$$

Solution. Setting $f(x)=0$, we have to solve all x such that

$$4(x-2)(x+3)(x-2021)^{19}=0.$$

Since we know that if for any a,b such that $ab=0$, then we can conclude that $a=0$ or $b=0$, we know that the x-values that satisfy the equation in the display are

$$x=2,-3,2021.$$

Example 2. Consider the polynmomial function

$$f(x)=x^3-6x^2+10x-8.$$

(a) State the degree and leading coefficient of the polynomial.
(b) Determine whether 4 is a zero/root of $f(x)$

See the footnote for the answers.[1]

We know some low-degree polynomials: Let a,b,c be constant numbers.

n (the degree)	General form of Degree n polynomial	Familiar Name
$n=0$	$f(x)=a$	constant function
$n=1$	$f(x)=ax+b$	non-constant lines (lines of nonzero slope)
$n=2$	$f(x)=ax^2+bx+c$	quadratics

Now the two most important questions are:

(1) Can we compute the all of the roots of a polynomial and how so?
(2) Is there relation between degree and the number of zeros a polynomial has?

[1](a) The degree is 3 and the leading coefficient is 1. (b) Plug in $x=4$ into $f(x)$ to find $f(4)=(4)^3-6(4)^2+10(4)-8=0$ so yes, it is a zero of $f(x)$.

Let's answer these questions for degrees $0, 1$ and 2 polynomials.

Let $f(x) = a$ be a constant. Setting $f(x) = 0$, we have $a = 0$ and so we have that a degree 0 polynomial can only have a root if it is the horizontal line $y = 0$.

Let $f(x) = ax + b$ (a line!). Setting $f(x) = 0$, we can solve for x:

$$ax + b = 0 \Longrightarrow x = -\frac{b}{a}$$

and so we have exactly 1 zero at $x = -\dfrac{b}{a}$.

Let's look at $f(x) = ax^2 + bx + c$. If we remember our algebra 2 well enough (if not, see the next section of notes), we have

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

and so we have up to two roots, depending on whether $b^2 - 4ac$ is positive, negative, or zero.

In summary, we have:

degree n	polynomial general form	location of zeros
0	$f(x) = a$	no zeros/roots unless $a = 0$.
1	$f(x) = ax + b$	$x = -\frac{b}{a}$
2	$f(x) = ax^2 + bx + c$	$x = \frac{-b+\sqrt{b^2-4ac}}{2a}, \frac{-b-\sqrt{b^2-4ac}}{2a}$.

Before we come to the takeaway of §4.1, we need to look at an important result, and a corollary (a consequence) of it that comes from mathematical induction (thanks POMI!)

Theorem 1 (Fundamental Theorem of Algebra, Gauss)**.** *Every non-constant polynomial with complex coefficients has at least one complex root.*

Proof. There are many proofs to this theorem. You can see them here:

`https://en.wikipedia.org/wiki/Fundamental_theorem_of_algebra`.

If you are interested in doing a project and some credit, let me know if you want to go through one of the proofs of the fundamental theorem of algebra. □

Some fun facts that won't matter unless you go into studying mathematics:

(1) there is a cubic and quartic formula but they are very long and disgusting looking. You can find their formulas here:
(2) There is no general formula for polynomials of degree 5 and greater. This result was proven by Galois and the proof of this makes for a very good project.

Corollary 2. *If $f(x)$ is a degree n polynomial with complex coefficients, i.e.,*

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

where each $a_i \in \mathbb{C}$ for $0 \leq i \leq n$, then we can find complex numbers $z_1, z_2, \ldots, z_n$ (not necessarily distinct) *such that the following equation holds:*

$$f(x) = a_n(x - z_1)(x - z_2)\cdots(x - z_n).$$

Definition. The total number of times any given root z_i appears in the factored form is called the **multiplicity** of the root $x = z_i$.

Example 3. Recall the **Example 1** where

$$f(x) = 4(x - 2)(x + 3)(x - 2021)^{19}.$$

The zeros are located at $x = 2, -3, 2021$, but the zeros $x = 2, -3$ each have multiplicity 1 while the root $x = 2021$ has multiplicity 19.

The entire purpose/lesson of §4.1 in the textbook can be summarised by this idea

Takeaway of §4.1

Let $f(x)$ be a polynomial of degree $n \geq 1$. Then

$$\begin{aligned} \deg(f) &\geq \text{number of zeros (counting multiplicity)} && \text{if } \mathrm{Dom}(f) = \mathbb{R} \\ \deg(f) &= \text{number of zeros (counting multiplicity)} && \text{if } \mathrm{Dom}(f) = \mathbb{C}. \end{aligned}$$

2. Solving Quadratics

Example 4. Find the roots (zeros) of the function

$$f(x) = x^2 - 9x + 18.$$

See footnote for answers[2]

The question is not answerable in a single immediate step, but requires us to do a little bit of work before we can arrive at the answer.

The entire purpose of the chapter is to ask us to convert polynomials

$$f(x) = ax^2 + bx + c \longrightarrow f(x) = k(x - \alpha)(x - \beta)$$

(where we need to find what α, β, k are equal to, provided that we have actual numbers for a, b, c) since when it is written as a bunch of product, as we saw from **Example 1**, we can just read off the zeros.

[2] $x = 6, 3$

We start off with quadratics since they are the first non-trivial case. There are three main algebraic methods for solving quadratics: factoring, completing the square, and the quadratic formula.

2.1. **Factoring.** Recall that

$$(x+a)(x+b) = x^2 + ax + bx + ab = x^2 + (a+b)x + ab$$

and so if we want to factor expressions of the form

$$x^2 + \alpha x + \beta$$

you simply need to compute numbers a and b such that

$$\begin{aligned} a + b &= \alpha \\ a \cdot b &= \beta \end{aligned}$$

Example 5. Try factoring the following expressions:

$$\begin{aligned} f(x) &= x^2 + 3x + 2 \\ g(x) &= x^2 + 9x + 20 \\ h(x) &= x^2 + 14x - 72 \end{aligned}$$

For $f(x)$, we need to find a and b such that

$$\begin{aligned} a + b &= 3 \\ a \cdot b &= 2 \end{aligned}$$

This gives us that $a, b = 1, 2$ and so

$$\boxed{f(x) = x^2 + 3x + 2 = (x+1)(x+2).}$$

For $g(x)$, we need to find a and b such that

$$\begin{aligned} a + b &= 9 \\ a \cdot b &= 20 \end{aligned}$$

This gives us that $a, b = 4, 5$ and so

$$\boxed{g(x) = x^2 + 9x + 20 = (x+4)(x+5).}$$

For $h(x)$, we need to find a and b such that

$$\begin{aligned} a + b &= 14 \\ a \cdot b &= -72 \end{aligned}$$

This gives us that $a, b = 18, -4$ and so

$$\boxed{h(x) = x^2 + 14x - 72 = (x+18)(x-4).}$$

If there is a $-x^2+\alpha x+\beta$ (as opposed to $x^2+\alpha x+\beta$) then factor the -1 out first, and then do the factoring methods above.

Example 6 (Factor out the gcd)**.** Factoring out

$$6x^2+9x \quad \text{and} \quad 4x^2+20x-56$$

can be made more easy by asking what is the common factor of each term?

Compute the gcd (greatest common divisor/greatest common factor) of all the terms for each problem:

$$\gcd\left(6x^2, 9x\right) = 3x$$
$$\gcd\left(4x^2, 20x, -56\right) = 4$$

and in each problem, let's factor out the gcd. This gives us

$$\begin{aligned} 6x^2+9x &= \underbrace{3x}_{\text{the gcd}}(2x+3) = \boxed{3x(2x+3)} \\ 4x^2+20x-56 &= \underbrace{4}_{\text{the gcd}}\left(x^2+5x-14\right) \\ &= \boxed{4(x-2)(x+7)} && \text{factoring } x^2+5x-14 \end{aligned}$$

What if we are left with trying to factor out a quadratic where the leading coefficient is not ± 1 (the term on the x^2 is not just ± 1?) Let's consider another example:

Example 7. Factor $f(x) = 2x^2+11x+12$.

(1) Compute ac: Since $a=2, b=11, c=12$, we have
$$ac = 2(12) = 24.$$
(2) Find the factors of ac that add up to b: We look at all the factors of 24:
$$\{1,24\}, \{2,12\}, \ \{3,8\}, \ \{4,6\}$$
and since $3+8=11$, we use that pair of numbers.
(3) After identifying the correct pair, rewrite b as the sum of the two pairs and factor:

$$\begin{aligned} f(x) &= 2x^2+11x+12 \\ &= 2x^2+(8x+3x)+12 && \text{rewrite } b \\ &= \left(2x^2+8x\right)+(3x+12) && \text{rearrange parantheses (associative property of } +) \\ &= 2x\,(x+4)+3(x+4) && \text{since } \gcd(2x^2,8x)=2x,\ \gcd(3x,12)=3 \\ &= (2x+3)(x+4) && \text{factoring out } (x+4) \end{aligned}$$

and so

$$\boxed{f(x) = (2x+3)(x+4)}$$

Example 8. Factor $f(x) = 2x^2-7x+6$.

(1) Compute ac: Since $a = 2, b = -7, c = 6$, we have

$$ac = 2(6) = 12.$$

(2) Find the factors of ac that add up to b: Since $b < 0$ with $c > 0$, our two factors are both negative numbers: so let's look at the negative pairs that multiply to 12 and add to -7:

$$\{-1, -12\}, \ \{-2, -6\}, \ \{-3, -4\}$$

and since $-3 + (-4) = -7$, we use that pair of numbers.

(3) After identifying the correct pair, rewrite b as the sum of the two pairs and factor:

$$\begin{aligned}
f(x) &= 2x^2 - 7x + 6 \\
&= 2x^2 + (-4x - 3x) + 6 && \text{rewrite } b \\
&= \left(2x^2 - 4x\right) + (-3x + 6) && \text{rearrange parantheses (associative property of } +) \\
&= 2x\,(x - 2) - 3(x - 2) && \text{since } \gcd(2x^2, -4x) = \pm 2x, \ \gcd(-3x, 6) = \pm 3 \\
&= (2x - 3)(x - 2) && \text{factoring out } (x - 2)
\end{aligned}$$

and so

$$\boxed{f(x) = (2x - 3)(x - 2)}$$

2.2. **Difference of Squares.** Expanding out the expression $(a + b)(a - b)$ gives us

$$\begin{aligned}
(a + b)(a - b) = a(a - b) + b(a - b) &= a^2 - ab + ba - b^2 \\
&= a^2 - b^2
\end{aligned}$$

That is, we have

$$\boxed{a^2 - b^2 = (a + b)(a - b) = (a - b)(a + b)}.$$

Example 9. Consider $p(x) = 25x^2 - 9$ and $q(x) = (4x - 9)(4x + 9)$. Factoring $p(x)$ using the difference of squares gives us

$$p(x) = \underbrace{25x^2}_{a^2} - \underbrace{9}_{b^2} = (a + b)(a - b) = (5x + 3)(5x - 3)$$

and expanding $q(x)$ gives us $a = 4x, b = 9$ so

$$q(x) = (4x - 9)(4x + 9) = (4x)^2 - (9)^2 = 16x^2 - 81.$$

Question 1. Without a calculator or pen/paper, compute

$$(999,999,999) \times (1,000,000,001) =$$

(Hint: use difference of squares)

2.3. **Completing the Square.** Completing the square is a method for rewriting quadratics into a nice looking formula:

$$x^2 + bx + c \rightsquigarrow (x+h)^2 + k$$

Expanding $(x+h)^2$ gives

$$(x+h)^2 = x^2 + 2hx + h^2$$

and so clearly, we have

$$(x+h)^2 + k = x^2 + 2hx + h^2 + k$$

and comparing this to the equation $x^2 + bx + c$, it is clear that we need

$$\begin{aligned} b &= 2h \\ c &= h^2 + k \end{aligned}$$

Example 10 (Page 214 example)**.** Find the roots of $f(x) = x^2 - 6x - 16$ by completing the square.

Remember that our goal is to compute the constants h, k such that

$$f(x) = x^2 - 6x - 16 = (x+h)^2 + k.$$

(1) Set $f(x) = 0$:

$$x^2 - 6x - 16 = 0.$$

(2) Identify what h must be equal to and compute h^2:

$$b = 2h = -6 \implies h = -3$$

and so computing h^2 gives us that $h^2 = 9$.

(3) Compute k using the equation $c = h^2 + k$:

$$\begin{aligned} c = -16 &= h^2 + k \\ &= 9 + k \\ &\implies k = -16 - 9 = -25 \end{aligned}$$

(4) rewrite the equation

$$\begin{aligned} f(x) &= x^2 - 6x - 16 \\ &= (x-3)^2 - 25 && \text{our work from above} \end{aligned}$$

(5) Compute the roots (solve for x)

$$\begin{aligned} (x-3)^2 - 25 = 0 &\implies (x-3)^2 = 25 \\ &\implies (x-3) = \sqrt{25} \\ &\implies (x-3) = \pm 5 \\ &\implies x = 3+5, 3-5 \end{aligned}$$

So our roots are at $\boxed{x = 8, -2.}$

- Note that to complete the square of $ax^2 + bx + c = 0$ that we need $a = 1$.

- In $x^2 + bx + c$ (since $a = 1$), the term with the x (your b) carries all of the important information.
  (1) Given $f(x) = x^2 + bx + c$, set $2h = b$ and solve for h.
  (2) Using the value of h, compute h^2.
  (3) Set $c = h^2 + k$ and solve for k.
  (4) celebrate! you've completed the square.

Example 11 (§4.2 #14)**.** Solve the equation by completing the square:

$$x^2 - 10x + 21 = 0.$$

Solution. Follow the steps above:

(1) Set $2h = b$, we have $b = -10$ so $h = -5$.
(2) Since $h = -5$, compute h^2 to be $h^2 = 25$.
(3) Set $c = h^2 + k$ and solve for k:

$$c = 21 = 25 + k \Longrightarrow k = -4$$

(4) The completed square form is

$$(x-5)^2 - 4 = 0.$$

Since we are asked to solve for x in the question, let's solve for x:

$$\begin{aligned}(x-5)^2 - 4 = 0 &\Longrightarrow (x-5)^2 = 4\\ &\Longrightarrow x - 5 = \sqrt{4}\\ &\Longrightarrow x - 5 = \pm 2\\ &\Longrightarrow x = 5 \pm 2\end{aligned}$$

so the final answers are $x = 5 - 2, 5 + 2$. So

$$\boxed{x = 3, 7.}$$

2.4. **Quadratic Formula.** The quadratic formula is a formula that will never fail to locate the zeros of a quadratic. It will solve any non-factorable or any non-complete-the-square-able quadratics.

Quadratic Formula

Given $ax^2 + bx + c = 0$, and $a \neq 0$, you can solve for x by using the formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The rest of this page only contains a proof of the quadratic formula; skip to next page if you are not interested.

Proof. The proof is reliant on completing the square. Since $a \neq 0$, it follows that

$$ax^2 + bx + c = a\underbrace{\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right)}_{\text{focus on this}}$$

and so if we set $ax^2 + bx + c = 0$, we have

$$a\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right) = 0 \Longrightarrow x^2 + \frac{b}{a}x + \frac{c}{a} = 0 \quad \text{(divide by } a\text{)}$$

and so if we start the process of completing the square:

(1) Set $2h = \dfrac{b}{a}$ and so we have $h = \dfrac{b}{2a}$.
(2) Compute the value of h^2 to be

$$h^2 = \left(\frac{b}{2a}\right)^2 = \frac{b}{2a} \cdot \frac{b}{2a} = \frac{b^2}{4a^2}.$$

(3) Setting $\dfrac{c}{a} = h^2 + k$, we compute k:

$$k = \frac{c}{a} - h^2 = \frac{c}{a} - \frac{b^2}{4a^2}.$$

(4) Rewrite the equation/the completed square form for $ax^2 + bx + c = 0$ is:

$$(x + \frac{b}{2a})^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0.$$

(5) Solve for x:

$$\begin{aligned}
(x + \frac{b}{2a})^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0 &\Longrightarrow (x + \frac{b}{2a})^2 = \frac{b^2}{4a^2} - \frac{c}{a} \\
&\Longrightarrow (x + \frac{b}{2a})^2 = \frac{b^2 - 4ac}{4a^2} \qquad \text{adding fractions} \\
&\Longrightarrow x + \frac{b}{2a} = \pm\sqrt{\frac{b^2 - 4ac}{4a^2}} \\
&\Longrightarrow x = -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\
&\Longrightarrow x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{\sqrt{4a^2}} \\
&\Longrightarrow x = -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
&\Longrightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}
\end{aligned}$$

Thus, this is how we arrive to the quadratic formula. □

The next page will be example computations.

Example 12. Consider the equation $f(x) = x^2 - x - 1$ and find the roots of $f(x) = 0$.

Since we are solving for $x^2 - x - 1 = 0$, we recognise that in the form $ax^2 + bx + c = 0$ that we need $a = 1, b = -1$ and $c = -1$. Plugging this into the quadratic formula gives us

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} = \frac{1 \pm \sqrt{5}}{2}$$

and so we have that the roots are

$$x = \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2}.$$

(The numbers above are called the golden ratio)

Example 13. Consider the equation $f(x) = x^2 - \sqrt{6}x + \frac{1}{2} = 0$ and find the roots of $f(x) = 0$.

It follows from the quadratic formula that after identifying that $a = 1, b = -\sqrt{6}, c = \frac{1}{2}$, we have that

$$\begin{aligned} x &= \frac{-(-\sqrt{6}) \pm \sqrt{(-\sqrt{6})^2 - 4(1)(\frac{1}{2})}}{2(1)} \\ &= \frac{\sqrt{6} \pm \sqrt{6 - 2}}{2} \\ &= \frac{\sqrt{6} \pm \sqrt{4}}{2} \\ &= \frac{\sqrt{6} \pm 2}{2} \end{aligned}$$

and so the roots are

$$x = \frac{\sqrt{6} + 2}{2}, \frac{\sqrt{6} - 2}{2}.$$

2.4.1. *The Discriminant.* Recall the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The expression inside the squareroot, the $b^2 - 4ac$, is called the **discriminant** and is often denoted with Δ. That is, the **discriminant** is

$$\Delta = b^2 - 4ac.$$

What it means:

The discriminant is useful since it tells us the number of real roots we our quadratic has. That is, if

$$f(x) = ax^2 + bx + c, \qquad a, b, c \in \mathbb{R}$$

then $\Delta = b^2 - 4ac$ and

$$\begin{aligned} \Delta > 0 &\iff \text{two distinct real roots} \\ \Delta = 0 &\iff \text{one real root with multiplicity 2} \\ \Delta < 0 &\iff \text{no real roots (two complex roots)} \end{aligned}$$

Example 14. Consider the equation $f(x) = 4x^2 + 5 - x$. Let's try to find the discriminant. We should remember $a = 4$, $b = -1$, and $c = 5$. The discriminant of f must then be

$$\Delta_f = b^2 - 4ac = (-1)^2 - 4(4)(5) = 1 - 80 < 0$$

so there are no real roots, but two complex roots.

3. Theorems to Help Us Locate Roots

Let $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ be a polynomial of degree $n \geq 1$ with *real* coefficients (i.e., $f \in \mathbb{R}[x]$ or $a_0, a_1, \ldots, a_n$ are real numbers), there are several results that help us restrict the possibilities when searching roots to the polynomial equation $f(x) = 0$.

Theorem 3 (Intermediate Value Theorem)**.** *Let $f(x)$ be a continuous function* (polynomials are continuous)*. If you can find real numbers a and b such that $f(a) > 0$ and $f(b) < 0$, then you know there must exist at least one x-value between a and b such that $f(x) = 0$.*

Example 15. Given that $f(x)$ is a function with the following table of values:

x	$f(x)$
-3	-49
-2	-12
-1	5
0	8

you know from the intermediate value theorem that $f(x) = 0$ must have a root somewhere in the interval $-2 < x < -1$ since $f(-1) > 0$ and $f(-2) < 0$.

Example 16 (§4.5 Exercise #18)**.** Is there a zero of $f(x) = 6x^3 + 24x^2 - 54x - 3$ between -6 and -5? Explain.

Evaluate $f(-6)$ and $f(-5)$ to find that $f(-6) = 753$ and $f(-5) = 717$ so the IVT does not give us any information other than that there may or may not be a zero between -6 and -5.

Theorem 4 (Complex Conjugate Roots Theorem)**.** *If the coefficients to the polynomial $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ are real, and if $x = u + iv$ is a root for some real numbers u, v, then the complex conjugate $x = u - iv$ is also a root.*

That is, complex roots always come in pairs.

Example 17.
- If $3 + 2021i$ is a root of the polynomial $f(x)$, then so must $3 - 2021i$.
- If $5 - \sqrt{7}i$ is a root of $f(x)$, then so will $5 + \sqrt{7}i$.
- If $3 + 2021i$ is a root of $f(x)$, then so will $3 - 2021i$.

Theorem 5 (Conjugate Radical Roots Theorem)**.** *If the coefficients to the polynomial $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ are rational numbers*[3]*, and if $x = u + v\sqrt{t}$, with $\sqrt{t}$ being irrational, then the conjugate radical $x = u - v\sqrt{t}$ is also a root.*

That is, irrational roots of rational polynomials always come in pairs.

- If $2 + 3\sqrt{6}$ is a root, then so is $2 - 3\sqrt{6}$
- If $\sqrt{7} - 2\sqrt{3}$ is a root, then so is $\sqrt{7} + 2\sqrt{3}$. Moreover, we also know that since
$$\sqrt{7} - 2\sqrt{3} = -2\sqrt{3} + \sqrt{7},$$
we know for a fact that $-2\sqrt{3} - \sqrt{7}$ is also a root and lastly, so must $-2\sqrt{3} + \sqrt{7}$ is also a root (total of 4 roots)
- If $\sqrt{3} - 2$ is a root of $f(x)$, then so will $-\sqrt{3} - 2$.

Theorem 6 (Rational Roots Theorem)**.** *If the coefficients to the polynomial $f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$ are integers, and if f has a rational root, then when the root is expressed in lowest possible terms, it will be of the form $x = \dfrac{u}{v}$, where u divides the constant term (a_0), and v divides the leading coefficient (a_n).*

Example 18.
- The numerator must divide the constant term and denominator divides leading term (see Section 5 for example) (**NCDL**)

Theorem 7 (Descartes' Rule of Signs)**.** *Let $f(x)$ be a polynomial with real coefficients, written in descending order of the variable exponent, i.e., write*
$$f(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0.$$
Then the number of positive roots of $f(x)$ is equal to the number of sign changes between consecutive (nonzero) coefficients, or is less than it by an even number.

Similarly, the number of negative roots of $f(x)$ is equal to the number of sign changes between consecutive (nonzero) coefficients of $f(-x)$, or is less than it by an even number.

Example 19. Consider the function
$$f(x) = x^7 + 6x^6 - 5x^5 + 2021x^4 - 2x^3 - x^2 + x - 9.$$

[3]note that $f \in \mathbb{Q}[x]$, not $\mathbb{R}[x]$. Also note that every integer is a rational number, so unless I give you a polynomial with an irrational coefficient such as π or $\sqrt{2}$, this theorem holds.

Then look at the signs of the coefficients (ignore coefficients that are equal to 0) after writing f in desecending order of exponents of x:

$$+, +, -, +, -, -, +, -$$

and so we see that there are 4 sign changes and so there must be either 4, 2, or 0 positive roots of $f(x)$.

To determine the number of negative roots of $f(x)$, plug in $f(-x)$ and see that

$$\begin{aligned} f(-x) &= (-x)^7 + 6(-x)^6 - 5(-x)^5 + 2021(-x)^4 - 2(-x)^3 - (-x)^2 + (-x) - 9 \\ &= -x^7 + 6x^6 + 5x^5 + 2021x^4 + 2x^3 - x^2 - x - 9 \end{aligned}$$

and so we see that the signs of the coefficients (in descending order of the exponents) are:

$$-, +, +, +, +, -, -, -$$

and so we see that there are 3 sign changes. Thus, there are either 3 or 1 negative roots of $f(x)$.

4. Polynomial Factors, Division

Let's consider a polynomial $p(x)$ and another polynomial $d(x)$ where $d(x)$ is *not* the zero-function. There is a division algorithm on polynomial spaces which says that **if we divide $p(x)$ by $d(x)$, we have the equation**

$$p(x) = q(x) \cdot d(x) + r(x)$$

for some (uniquely determined) polynomials $q(x)$ and $r(x)$ where $\deg(r) = 0$ or $\deg(r) < \deg(d)$.

The polynomial $q(x)$ is the *quotient* and the $r(x)$ is called the *remainder*.

In particular, if we assume that $\deg(p) \geq 1$, and let $d(x) = (x - k)$ for some constant k, this division algorithm gives us

$$p(x) = q(x) \cdot (x - k) + r$$

where r is a constant. Setting $x = k$ in this equation gives us that $p(k) = r$, which says that when $p(x)$ is divided by $(x - k)$, the remainder is exactly $p(k)$. This is called the *remainder theorem.*

Thus, we conclude with the *factor theorem* (a consequence of the remainder theorem):

Theorem 8 (Factor Theorem)**.** *The constant $x = k$ is a root to the real polynomial $p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n = 0$ if and only if $(x - k)$ is a factor of $p(x)$.*

Takeaway of Factor Theorem

Given some polynomial $p(x)$, if you can find a constant $x = k$ such that when you synthetically divide $p(x)$ by k (or divide by $(x-k)$ in long division) and get a remainder 0, then $p(x)$ has a root as $x = k$ and you only need to work on finding the roots of the quotient!

5. Graphing a Polynomial

(1) Identify y-intercept by evaluating $f(0)$
(2) Identify the x-intercepts by solving the polynomial equation $f(x) = 0$ by
   - methods of factoring
   - testing possible roots using synthetic division (the Factor Theorem says if you get remainder $= 0$, then that number you tested is a root!) and reducing the polynomial to the quotient (keep reducing until you get a quadratic; then use quadratic formula)
   - use intermediate value theorem, rational roots theorem, complex conjugate roots, and conjugate radical roots theorem, Descartes' rule of signs to help identify roots to test
(3) After finding the $z_1, \ldots, z_n$ such that $f(x) = a_n(x - z_1) \cdots (x - z_n)$, determine the multiplicity of each root
   - if the multiplicity is even, then the graph will bounce off the x-axis at the root
   - if the multiplicity is odd, then the graph will cross the x-axis at the root
(4) Create a table of values, where we need to determine the sign of $f(x)$ at each x-intercept.
(5) Graph it!

Example 20. Following the steps above, graph the polynomial $f(x) = x^4 - 9x^2 - 4x + 12$.

Solution.

(1) Identify y-intercept: Plugging in $x = 0$ into $f(x)$, we get

$$f(0) = 12$$

so the y-intercept for $f(x)$ is at $(0, 12)$.

(2) Identify the x-intercepts by solving the polynomial equation $f(x) = 0$:

This is the longest step in the process. Using the rational roots theorem, we know that we need to have a root $x = \frac{u}{v}$ where u divides 12 and v divides 1 so the possible roots to test are:

$$x = \pm\frac{1}{1}, \pm\frac{2}{1}, \pm\frac{3}{1}, \pm\frac{4}{1}, \pm\frac{6}{1}, \pm\frac{12}{1}.$$

Try testing $(x - 1) = 0$ or equivalently $x = 1$ by synthetic division:

1	1	0	-9	-4	12

$$\begin{array}{ccccc} 0 & 1 & 1 & -8 & -12 \\ \hline 1 & 1 & -8 & -12 & 0 \end{array}$$

and since the remainder is zero, we know that

$$f(x) = (x-1)(x^3 + x^2 - 8x - 12)$$

and so we now need to find the roots of the equation $x^3 + x^2 - 8x - 12 = 0$.

Using the list of candidates $x = \pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$ provided by the rational roots theorem, we test $(x+2)$, or equivalently $x = -2$ by synthetic division:

$$\begin{array}{r|rrrr} -2 & 1 & 1 & -8 & -12 \\ & 0 & -2 & 2 & 12 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

and since we see that the remainder is 0, we conclude that $x = -2$ is a root and so we know that

$$x^3 + x^2 - 8x - 12 = (x+2)(x^2 - x - 6)$$

However, you can solve the quadratic equation

$$x^2 - x - 6 = 0 \Longrightarrow (x-3)(x+2) = 0 \quad \text{(using any method you want)}$$

to find that $x = 3, -2$ are the other roots. That is, we have that

$$\begin{aligned} f(x) &= x^4 - 9x^2 - 4x + 12 \\ &= (x-1)(x^3 + x^2 - 8x - 12) \\ &= (x-1)(x+2)(x^2 - x - 6) \\ &= (x-1)(x+2)(x+2)(x-3) \\ &= (x-1)(x+2)^2(x-3) \end{aligned}$$

The x-intercepts are at $x = 1, -2, 3$.

(3) Determine the multiplicity of each root: We have that the root at $x = 1$ has multiplicity 1, the root at $x = -2$ has multiplicity 2, and the root $x = 3$ has multiplicity 1.
- the graph of $f(x)$ will bounce off the x-axis at $x = -2$ (even multiplicity)
- the graph of $f(x)$ will cross the x-axis at $x = 1, 3$ (odd multiplicity)

(4) Create a table of values, where we need to determine the sign of $f(x)$ at each x-intercept.

Since the x-intercepts are at $x = -2, 1, 3$ (rewriting it from smallest to greatest), we split up the interval $(-\infty, \infty)$ by the x-intercepts:

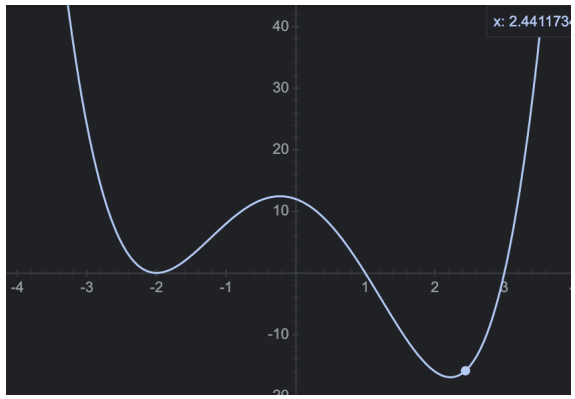
interval :	$(-\infty, -2)$	$(-2, 1)$	$(1, 3)$	$(3, \infty)$
$x =$ test point	-3	0	2	4
$f(x)$	$f(-3) = 24$	$f(0) = 12$	$f(2) = -16$	$f(4) = 108$
sign :	$+$	$+$	$-$	$+$

(5) Graph it!

Using the information that you have above, you know for a fact that:

- $f(x)$ is continuous (since $f(x)$ is a polynomial)
- The graph of $f(x)$ stays above the x-axis on the interval $(-\infty, -2)$ and on the interval $(-2, 1)$, kissing the x-axis at $x = -2$.
- The graph of $f(x)$ crosses the x-axis at $x = 1$, going from positive to negative
- The graph of $f(x)$ is below the x-axis on the interval $(1, 3)$ and crosses the x-axis at $x = 3$, going from negative to positive
- $f(0) = 12$ and so there is a y-intercept at 12.

You should get something like this:



- When doing this on your own, you don't need to have the precise values of the (local) maximum or minimum, unless a graphing calculator is available (introductory calculus will give you the necessary tools to determine these values without a calculator).